Prompt 4 - no MCP tool description - run 2 (seed 42)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](file://./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](file://./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

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🧪 Compound Activity Prediction & Explainability Analysis

I have successfully trained two Graph Neural Network (GNN) models—**GCN** and **GAT**—on the `chembl_activity_data_P14416_binary` dataset to predict compound activity. Below is a comprehensive comparison of their performances, followed by an explainability analysis using EdgeSHAPer.

📊 Model Performance Comparison

Metric	GCN Model	GAT Model
Test Accuracy	91.62%	91.50%
Balanced Accuracy	91.81%	91.73%
MCC	0.838	0.839
AUC-ROC	0.970	0.975
Precision	0.869	0.859
Recall	0.973	0.985

Both models exhibit excellent predictive performance with high AUC-ROC scores (>0.97). The GAT model slightly outperforms GCN in terms of AUC-ROC and Recall, while GCN shows marginally better Precision.

🔍 EdgeSHAPer Explainability Analysis

To understand how these models arrive at their predictions, I selected a compound from the test set that was **correctly predicted as active** by both models with high confidence:

- **Compound SMILES:** `O=C(NC1CCC(CCN2CCC(c3cccc4c3OCO4)CC2)CC1)c1ccc(Cl)cc1`

- **True Label:** 1 (Active)
- **GCN Confidence:** 0.989
- **GAT Confidence:** 0.999

📈 Explainability Metrics Comparison

Explainability Metric	GCN Model	GAT Model
Fidelity	0.664	**0.957**
Trustworthiness	0.586	**0.770**
Infidelity	0.476	0.356

💡 Key Differences in Learning Characteristics

1. **Interpretability & Fidelity:** The **GAT model demonstrates significantly higher fidelity (0.957 vs. 0.664)** and trustworthiness (0.770 vs. 0.586). This indicates that GAT's attention mechanism allows it to more accurately pinpoint the specific bonds and structural motifs responsible for its prediction compared to GCN's simpler aggregation approach.
2. **Bond Contribution Patterns:**
   - **GCN** relies on a broader set of bonds with more uniformly distributed contributions, reflecting its message-passing nature where information is aggregated from all neighbors.
   - **GAT** exhibits a more "sharp" attention profile, highlighting a smaller subset of critical bonds with higher weights. This aligns with GAT's ability to learn which neighbors are most important for the current node's representation.
3. **Structural Motifs:** Both models correctly identify the central core and key functional groups (like the chlorine-substituted phenyl ring and the furan-like oxygen-containing ring) as highly relevant. However, GAT assigns higher importance scores to the bonds connecting the core scaffold to these active pharmacophores, suggesting it has learned a more discriminative structural representation.

📝 Conclusion

While both GCN and GAT achieve near-identical predictive accuracy on this task, **GAT provides superior explainability**. Its higher fidelity and trustworthiness scores in EdgeSHAPer analysis suggest that it captures the underlying structure-activity relationships more precisely, making it a more reliable choice when model interpretability is crucial for medicinal chemistry applications. The attention weights in GAT offer chemists clearer insights into which specific molecular interactions drive the predicted activity.